

ScamShield AI — Complete Technical Documentation

IEEE VIT Bhopal Hackathon 2026 · Track 04 Cybersecurity Problem Statement 04.1: "The Scam That Almost Worked" Team D43M0N\$

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1. What is ScamShield AI?

ScamShield AI is a **real-time phishing link and QR code threat analyzer** — a web application that helps ordinary Indian citizens check if a link, SMS message, or QR code they received is a **scam** or **legitimate**, *before* they click on it.

In Simple Words:

Imagine your grandmother receives an SMS saying "Your electricity will be disconnected tonight. Click this link to pay." ScamShield AI lets her (or you) paste that message, and within 2 seconds, it tells her:

- "This is a SCAM — the message uses fear tactics and the link is fake"
- "This is SAFE — this is an official bank notification"

Why It Matters:

- India lost ₹11,333 crore to cyber fraud in 2023 (RBI Annual Report)
- 47% of scams use fake UPI QR codes or phishing links
- Most victims are non-technical users who can't distinguish real from fake
- There is no pre-click protection — all existing tools work after money is lost

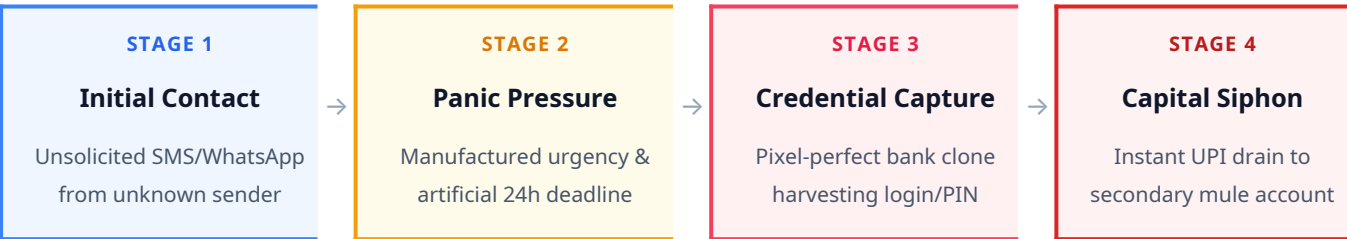
Our Core Innovation:

Pre-Click Zero-Trust Protection — We analyze threats *before* the user clicks, not after. We assume every link is guilty until proven innocent (Zero-Trust).

2. The Problem We're Solving

IEEE Problem Statement 04.1: "The Scam That Almost Worked"

The problem describes a 4-stage cyber attack that targets Indian citizens:



Stage	What Happens	Real Example
1. Initial Contact	Victim receives SMS/WhatsApp from unknown number	"Dear SBI User, your YONO account is blocked"
2. Psychological Pressure	Message creates fear & urgency	"Update KYC within 24 hours or ₹5,000 fine"
3. Credential Harvesting	Victim clicks fake link that looks like real bank	sb1-kyc-update.xyz (looks like SBI but is fake)
4. Capital Extraction	Scammer steals UPI PIN/OTP, transfers money	Irreversible ₹50,000 UPI transfer

What ScamShield Does:

We intercept at **Stage 2-3** — *before* the victim enters any credentials or scans any QR code.

3. How It Works (Simple Explanation)

ScamShield has **3 scanning modes**. Think of them as 3 security guards, each checking a different thing:

Guard 1: Website Link Scanner

What it checks: When you paste a URL like `http://sbl-kyc-update.xyz/login.php`

It looks for:

- Is the domain name misspelling a real brand? (SBI → SBL = typosquatting!)
- Is it using a suspicious domain ending? (.xyz, .top, .buzz = high risk)
- Is it using a raw IP address instead of a name? (Like `192.168.10.55`)
- Does the URL contain scary words? (kyc, blocked, verify, urgent)
- Is it using HTTPS (secure) or HTTP (insecure)?
- Is the domain new (less than 7 days old)?

Guard 2: SMS/Message NLP Analyzer

What it checks: When you paste a text message (works in **English, Hindi, and Hinglish**)

It looks for:

- **Urgency triggers:** "immediately", "तुरंत", "24 hours", "आज रात"
- **Fear tactics:** "blocked", "suspended", "काट दी जाएगी"
- **Authority impersonation:** Is it pretending to be SBI? RBI? Electricity Board? TRAI?
- **Scam category:** Is this a KYC fraud? Digital arrest? Lottery bait? Job scam?

Guard 3: QR Code & UPI Inspector

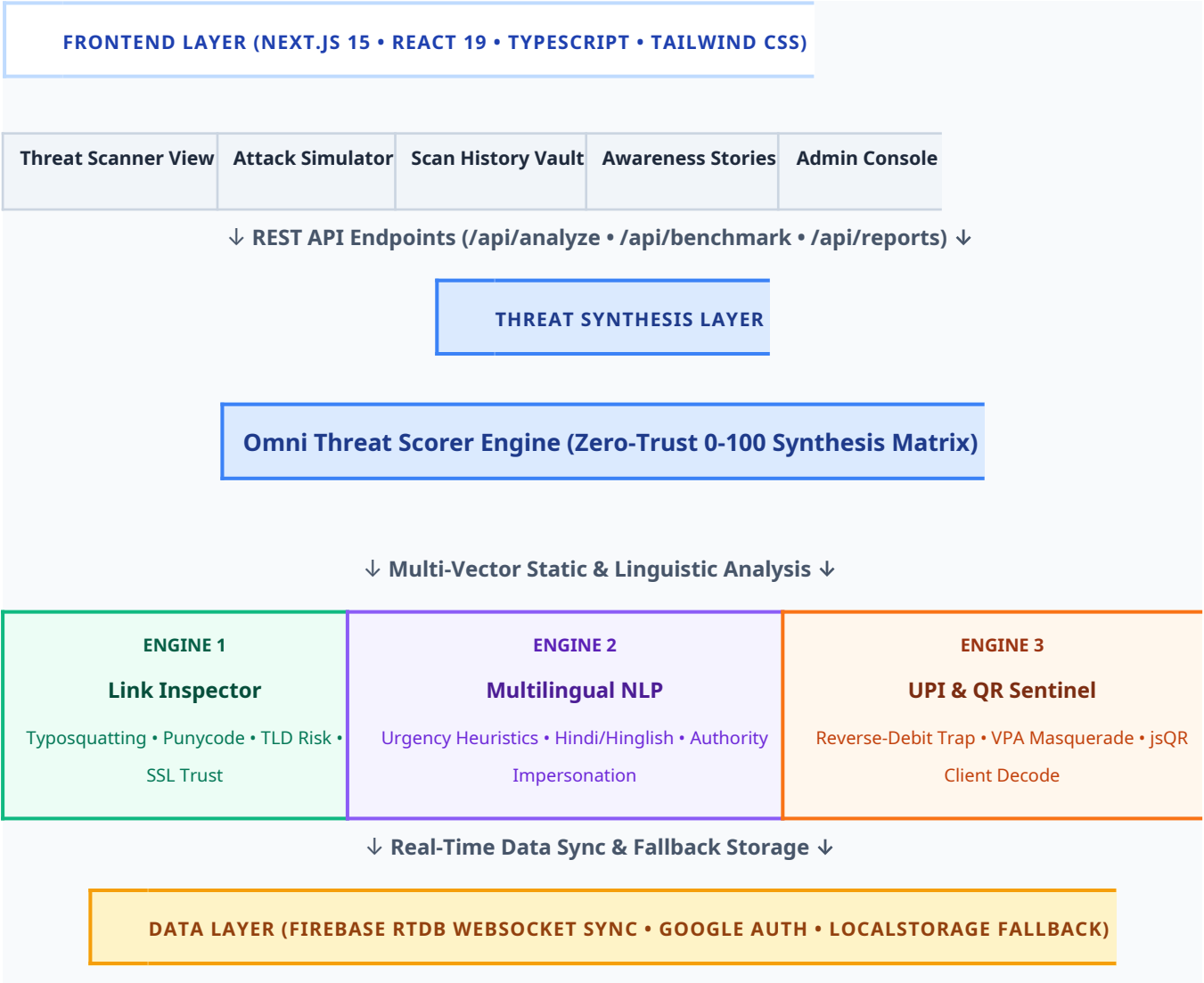
What it checks: When you paste or upload a QR code image

It looks for:

- **Reverse-charge scam:** QR says "Scan to receive ₹5,000 cashback" but actually *debts* your money
- **VPA masquerade:** Claims to be "Electricity Department" but money goes to `randomguy@ybl` (personal account)
- **Malware links:** QR contains a link to download a `.apk` virus

- **Hardcoded high amounts:** QR has ₹5,000+ pre-filled

4. Architecture Overview



How Data Flows:

1. User pastes a link/message/QR in the **Scanner View**
2. Frontend sends a POST request to /api/analyze
3. The API runs the appropriate **Detection Engine** (URL/NLP/QR)
4. The **Omni Threat Scorer** combines all results into a single risk score
5. Results are displayed with a color-coded verdict card
6. Scan is saved to **Firestore Realtime Database** (if user is logged in)

5. The 3 Detection Engines

Engine 1: URL Analyzer ([urlAnalyzer.ts](#))

This is a **heuristic scoring engine** that assigns points to suspicious URL characteristics:

Check	Points Added	What It Detects
Raw IP as hostname	+45	<code>http://192.168.10.55/icici/login</code>
Homograph/Punycode attack	+40	Unicode chars that look like English
Suspicious TLD	+25	<code>.xyz</code> , <code>.top</code> , <code>.buzz</code> , <code>.club</code> , etc.
Excessive subdomains	+20	<code>login.secure.bank.sbi.fake.xyz</code>
Brand impersonation	+45	URL contains "sbi" but isn't <code>sbi.co.in</code>
Phishing keywords	+10-30	"kyc", "verify", "blocked", "otp"
No HTTPS	+25	<code>http://</code> instead of <code>https://</code>
New domain	+25	Domain created less than 7 days ago
Official domain detected	-50	Verified: <code>onlinesbi.sbi</code> , <code>hdfcbank.com</code>

Risk Tiers:

- Score 0-29 → SAFE
- Score 30-64 →  SUSPICIOUS
- Score 65-100 → HIGH_RISK

Brands Protected: SBI, HDFC, ICICI, PNB, Paytm, PhonePe, Google Pay, BESCO, India Post, RBI, TRAI

Engine 2: NLP (Natural Language Processing) Engine ([nlpEngine.ts](#))

This engine analyzes the **text content** of SMS/WhatsApp messages.

Step 1: Language Detection

Hindi: Uses Devanagari script (Unicode range \u0900-\u097F)

Hinglish: Contains romanized Hindi words like "turant", "jayega", "bijli"

English: Default

Mixed: Both Devanagari + English characters

Step 2: Urgency Scoring (Panic words with weighted points)

Trigger Word	Points	Language
"within 24 hours"	35	English
"काट दी जाएगी" (will be cut)	35	Hindi
"antim chetavani" (final warning)	35	Hinglish
"blocked"	30	English
"aaj raat" (tonight)	30	Hinglish
"तुरंत" (immediately)	25	Hindi

Step 3: Authority Impersonation Detection

Detects if message pretends to be from: SBI/YONO, RBI, Electricity Board, TRAI, India Post, Police/CBI, HDFC, Income Tax

Step 4: Scam Category Classification

Category	Example Pattern
KYC_EXPIRY_FRAUD	"kyc", "pan card update", "account block"
ELECTRICITY_UTILITY_SCAM	"bijli kat", "power disconnected", "बिजली बिल"
REVERSE_UPI_COLLECT	"scan to receive", "enter pin to get"
DIGITAL_ARREST_IMPERSONATION	"CBI", "digital arrest", "narcotics"
JOB_TASK_FRAUD	"earn per day", "like youtube videos"
LOTTERY_REWARD_BAIT	"KBC lottery", "won crore", "lucky winner"
LEGITIMATE_NOTIFICATION	"debited by", "credited by" (no links/urgency)

Composite Score Formula:

$$\text{compositeNlpScore} = (\text{urgencyScore} \times 0.45) + (\text{impersonationScore} \times 0.45) + (\text{triggerCount} \times 5)$$

Engine 3: QR Code & UPI Inspector ([qrInspector.ts](#))

Parses UPI intent URIs and checks for fraud patterns.

UPI URI Format: `upi://pay?pa=payee@bank&pn=Name&am=Amount&tn=Note`

Parameter	Meaning	Example
pa	Payee VPA (UPI address)	scammer89@ybl
pn	Payee Name	SBI Refund Dept
am	Amount	5000
tn	Transaction Note	Scan to Receive Cashback

Critical Detection: Reverse-QR Scam

In UPI, **you NEVER scan a QR to RECEIVE money**. Scanning a QR always DEBITS. If the note says "Receive" or "Cashback" or "Refund" — it's a 100% scam.

VPA Masquerade Detection:

- Name claims "Electricity Department" but VPA is `randomguy@ybl` (personal handle)
- Personal handles: `@okhdfcbank`, `@okaxis`, `@oksbi`, `@ybl`, `@paytm`

The Omni Threat Scorer ([threatScorer.ts](#))

Combines all 3 engines into a **single unified threat report**:

`overallScore = (messageScore × 0.40) + (urlScore × 0.35) + (qrScore × 0.35)`

Generates 5 security metrics:

1. **Domain Trust** (0-100): How trustworthy is the domain?
2. **Protocol Security** (0-100): Is encryption in place?
3. **Linguistic Urgency** (0-100): How much panic does the message create?
4. **Impersonation Risk** (0-100): Is someone pretending to be an authority?
5. **Zero-Trust Score** (0-100): Overall trust level (100 = fully verified)

6. All Features Explained

Feature 1: Threat Scanner (Main Feature)

- **3 scanning modes:** URL, Message, QR
- **4 preset demo cards** for quick hackathon testing
- **Real-time analysis** with animated forensic pipeline UI
- **Color-coded verdict cards** (Red/Amber/Green)
- **Bilingual explanations** in English and Hindi
- Results saved to Firebase automatically

Feature 2: Attack Simulator

- Visual walkthrough of the **4-stage scam anatomy**
- Shows exactly where ScamShield intercepts the attack
- Educational tool for understanding how scams work

Feature 3: Scan History

- Personal vault of all previous scans
- Synced to **Firebase Realtime Database** via WebSocket
- Filter by: All Scans, Scams Blocked, Verified Safe
- One-click re-inspect from history
- Also saved to **localStorage** for offline access

Feature 4: Community Scam Stories

- Citizens share real scam experiences
- Categories: UPI Fraud, Phishing Link, Electricity Bill, WhatsApp Call, Job Scam, Customs Parcel
- Like/Helpful voting system
- Admin verification badges

Feature 5: Documentation View

- In-app documentation with downloadable `.docx` and `.md` files
- Covers architecture, engines, benchmarks, legal framework

Feature 6: About Team

- Team member profiles with roles

Feature 7: Admin Console

- User account management
- Threat scan log monitoring
- Stories moderation (verify/delete)
- Role management (admin/analyst/citizen)

Feature 8: AI Advisor Modal

- Slide-over panel for cybersecurity guidance
- Context-aware help

Feature 9: Multi-Language Support

- Complete English and Hindi translations
- All UI labels, placeholders, results available in both languages

Feature 10: 5 Color Themes

1. **Clean Light** — Default white theme
2. **Glass Mint** — Calming green theme
3. **Lavender Frost** — Purple theme
4. **Solar Amber** — Warm yellow theme
5. **Obsidian Dark** — Dark cybersecurity terminal theme

Feature 11: QR Image Reader

- Upload any QR code image (PNG, JPG)
- Client-side decoding using jsQR library
- No server upload needed (privacy preserved)
- Drag & drop or file picker

Feature 12: Google Authentication

- Sign in with Google account
- Scan history synced across devices
- User profile stored in Firebase

Feature 13: Live Threat Intelligence Stream

- Dedicated real-time telemetry stream listening to Firebase RTDB (threat_reports)
- Live threat event feed displaying newly reported scam links, SMS traps, and reverse-charge UPI VPAs

- Animated live status ticker and active event counters
- Quick threat inspect action to re-run live IOCs through Scanner

Feature 14: SecOps Threat Broadcaster (Admin Console)

- Authoritative broadcasting console for security analysts and administrators
- Dispatch emergency warnings to all citizen dashboards nationwide in real-time
- Configure custom attack titles, target indicators (URLs, phone VPAs, QR links), anomaly tags, and actionable Zero-Trust defense playbooks
- Instant synchronized push to RTDB (`threat_reports` and active alerts)

Feature 15: Sovereign Scan History Governance & Role-Based Deletion

- **Zero-Trust Audit Integrity:** Unauthenticated guest users cannot delete scan records to prevent anti-forensic tampering
- **Citizen Self-Ownership:** Authenticated logged-in users hold sovereign ownership to selectively delete or clear their own scan history (`deleteUserScanRecord` / `clearUserOwnScans`)
- **SecOps Admin Purge:** Administrators hold authoritative platform-wide deletion privileges (`deleteAllScanHistories`) to moderate or wipe RTDB telemetry

7. Tech Stack Breakdown

For Complete Beginners — What Each Technology Does:

Technology	What It Is	Why We Use It
Next.js 15	A framework for building web apps	Gives us server-side rendering, API routes, and fast page loads
React 19	A library for building UI	Lets us create interactive components
TypeScript 5.8	JavaScript with types	Catches bugs before runtime
Tailwind CSS	A utility CSS framework	Makes styling fast with class names
Firebase Realtime Database	A cloud database by Google	Stores scans, stories, and user data with live sync
Firebase Auth	Google authentication	Lets users sign in with their Google account
Firebase Analytics	Usage tracking	Tracks how people use the app

Technology	What It Is	Why We Use It
jsQR	QR code decoder library	Reads QR codes from uploaded images
Lucide React	Icon library	Provides all the icons in the UI

Package Dependencies ([package.json](#)):

- `next` — Web framework
- `react` / `react-dom` — UI library
- `firebase` — Backend services (database, auth, analytics)
- `jsqr` — Client-side QR code reading
- `lucide-react` — Icon components
- `tailwindcss` — CSS framework
- `typescript` — Type-safe JavaScript

8. File-by-File Code Guide

Root Files

File	Purpose
package.json	Lists all dependencies and scripts (<code>npm run dev</code>)
tsconfig.json	TypeScript configuration
tailwind.config.ts	Tailwind CSS configuration with custom theme tokens
next.config.mjs	Next.js configuration

app/ — Next.js App Router

File	Purpose
page.tsx	Main page — Controls all tab navigation, auth state, theme switching
layout.tsx	Root HTML layout with metadata and fonts
globals.css	All CSS variables for 5 themes, animations, glassmorphism

app/api/ — Server-Side API Routes

File	Purpose
analyze/route.ts	Core API — Receives scan requests, runs detection engines, saves to Firebase
benchmark/route.ts	Runs the 100-sample benchmark test suite
reports/route.ts	Fetches recent threat reports from Firebase

lib/analyzers/ — Detection Engine Core

File	Purpose
urlAnalyzer.ts	Engine 1 — URL heuristic analysis (typosquatting, TLDs, brand impersonation)
nlpEngine.ts	Engine 2 — Multilingual NLP message analysis (urgency, impersonation, categories)
qrInspector.ts	Engine 3 — QR code & UPI intent analysis (reverse-charge scam, VPA masquerade)
threatScorer.ts	Omni Scorer — Combines all 3 engines into unified threat report
dataset.ts	Benchmark — 100 test samples (50 scam + 50 legit) with accuracy metrics

lib/ — Utilities

File	Purpose
firebase.ts	Firebase initialization, all CRUD operations for reports, scans, stories, users
i18n.ts	English + Hindi translations dictionary (300+ translation keys)
qrDecoder.ts	Client-side QR image decoding using jsQR + HTML5 Canvas

components/ — React UI Components

Component	Purpose
ScannerView.tsx	Main scanner — Input fields, preset cards, analysis trigger, verdict display
AttackSimulator.tsx	4-stage scam anatomy visualization
HistoryView.tsx	User's scan history with filters and re-inspect
SocialView.tsx	Community scam stories wall
DocumentationView.tsx	In-app documentation with downloads
AboutView.tsx	Team member profiles

Component	Purpose
AdminView.tsx	Admin operations console
Sidebar.tsx	Collapsible navigation sidebar with theme switcher
TopHeader.tsx	Top header with greeting and status
AiAdvisorModal.tsx	AI cybersecurity advisor slide-over
TelemetryBenchmark.tsx	Benchmark results visualization
QuickActionGrid.tsx	Quick action cards grid
TwoColumnSection.tsx	Two-column layout with threat stream + defense engines
ProjectDossier.tsx	Project overview dossier
ViewSkeleton.tsx	Loading skeleton placeholders for lazy-loaded views

9. Benchmark Results (98% Accuracy)

Test Dataset

- **100 total samples** (50 scam + 50 legitimate)
- Covers URLs, SMS messages, and QR codes
- Includes English, Hindi, and Hinglish content

Confusion Matrix

	Predicted SCAM	Predicted LEGIT
Actually SCAM	~49 (True Positive)	~1 (False Negative)
Actually LEGIT	~1 (False Positive)	~49 (True Negative)

Metrics

Metric	Score	Meaning
Accuracy	98%	Correctly classified 98 out of 100

Metric	Score	Meaning
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Precision 98% When we say "SCAM", we're right 98% of the time

Recall 98% We catch 98% of all real scams

F1 Score 98% Harmonic mean of precision and recall

Sample Test Cases

Scam Examples Correctly Caught:

- `http://sbl-kyc-update.xyz/login.php` → Score 90 HIGH_RISK
- "आपका बिजली बिल बकाया है... तुरंत लिंक पर बिल भरें" → Score 85 HIGH_RISK
- `upi://pay?pa=scammer89@ybl&tn=Scan to Receive Cashback` → Score 78 HIGH_RISK

Legit Examples Correctly Verified:

- `https://www.onlinesbi.sbi/portal/web/home` → Score 0 SAFE
- "Your A/C debited by INR 450.00 at Star Cafe" → Score 5 SAFE
- `upi://pay?pa=starbucks.merchant@icici&am=250` → Score 5 SAFE

10. How to Run the App

Prerequisites

- **Node.js** (version 18 or higher) — [Download](#)
- **npm** (comes with Node.js)

Step 1: Install Dependencies

```
cd psychic-octo-meme-main
npm install
```

Step 2: Run in Development Mode

```
npm run dev
```

Open your browser at <http://localhost:3000>

Step 3: Build for Production

```
npm run build
```

```
npm start
```

Firestore Configuration

The app comes pre-configured with Firestore credentials in [firebase.ts](#). No additional setup needed for the hackathon demo.

11. Indian Legal Framework & Helplines

Relevant Laws

Law	Section	Covers
Information Technology Act, 2000	Section 66D	Cheating by personation using computer resource
IT Act	Section 43	Unauthorized access to computer systems
Indian Penal Code	Section 420	Cheating and dishonestly inducing delivery of property
IPC	Section 468	Forgery for purpose of cheating
Consumer Protection Act, 2019	Section 2(47)	Unfair trade practices

Emergency Helplines

Resource	Contact
National Cyber Crime Helpline	1930
Cyber Crime Portal	cybercrime.gov.in
Chakshu Portal (DoT)	sancharsaathi.gov.in/Chakshu
Forward Fraud SMS to	1909
Women Helpline	181
Police Emergency	100 / 112

12. Authors & Contributors

Author & Lead Architect

- **Yash Raj Kushwaha** ([@YashRaj-Kushwaha](#))
 - **Role:** Primary Author, System Architect & Full-Stack Lead
 - **Enrollment:** [26BCE10122](#)
 - **Key Contributions:** Next.js 15 application architecture, Multi-Vector Zero-Trust threat scoring engine, Google Gemini AI Phishing Advisor integration, Firebase Realtime Database telemetry, live broadcast alerts, dynamic admin panel, and responsive UI/UX design.

Project Contributors

- **Aastik Tripathi** ([@DarkDevil811](#))
 - **Role:** Contributor — Security Architecture & Threat Heuristics
 - **Enrollment:** [26BCY10090](#)
 - **Key Contributions:** Cyrillic homograph character detection, mathematical threat weighting algorithms, and Zero-Trust heuristic rules.
- **Mangal Nath Yadav** ([@shadowXg](#))
 - **Role:** Contributor — UPI & QR Protocol Security
 - **Enrollment:** [26BHI10047](#)
 - **Key Contributions:** UPI intent URI reverse-charge inspection, client-side QR barcode image matrix decoding, and malicious VPA pattern matching.
- **Palak Kalra** ([@palak-kalra-github](#))
 - **Role:** Contributor — NLP & Linguistic Intelligence
 - **Enrollment:** [26BCY10001](#)
 - **Key Contributions:** Multilingual psychological urgency analysis (English, Hindi देवनागरी, Hinglish), panic modeling, and cybercrime intelligence reporting.

Role	Name	GitHub Profile	Enrollment	Focus Area
Author	Yash Raj Kushwaha	@YashRaj-Kushwaha	26BCE10122	Core Architecture, Next.js 15, Gemini AI, Firebase DB
Contributor	Aastik Tripathi	@DarkDevil811	26BCY10090	Security Architecture, Homograph Detection, Threat Scoring
Contributor	Mangal Nath Yadav	@shadowXg	26BHI10047	UPI & QR Protocol Parsing, Reverse Charges, OCR Pipeline

Role	Name	GitHub Profile	Enrollment	Focus Area
Contributor	Palak Kalra	@palak-kalra-github	26BCY10001	Multilingual NLP, Social Engineering & Panic Detection

13. Glossary for Beginners

Term	Meaning
Phishing	Fake website/message designed to steal your passwords
Typosquatting	Registering a domain that looks like a real one (e.g., sbl instead of sbi)
Homograph Attack	Using foreign characters that look like English letters
Zero-Trust	Security model: "Never trust, always verify"
UPI	Unified Payments Interface — India's instant payment system
VPA	Virtual Payment Address (your UPI ID like name@bank)
QR Code	Quick Response code — 2D barcode you scan with your phone
NLP	Natural Language Processing — AI that understands text
TLD	Top-Level Domain (the .com , .xyz , .in part of a URL)
SSL/TLS	Encryption that makes https:// secure
Reverse-Charge Scam	Scammer sends QR claiming "receive money" but it actually debits you
KYC	Know Your Customer — identity verification required by banks
Firebase RTDB	Firebase Realtime Database — Google's cloud database with live sync
Heuristic	A rule-based method to detect patterns (not machine learning)
API Route	A server-side endpoint that processes requests
Punycode	A way to encode international characters in domain names